# Geometry Review: Unit 3 — Triangles & Congruence

Answer Key

## Station 1: Classifying Triangles and the Triangle Inequality

1. Isosceles
2. Scalene
3. (a) right; (b) acute; (c) obtuse
4. Yes; $4 + 7 = 11 > 10$ (and the other sums are larger)
5. No; $3 + 5 = 8 < 9$
6. $5 < x < 21$
7. $0 < x < 12$
8. 22 *sides* $9, 9, 4$ *(since* $4 + 4 < 9$*)*
9. $BC < AC < AB$
10. $60°$

## Station 2: Proving Triangles Congruent

1. SSS
2. SAS
3. ASA
4. AAS
5. HL
6. Not enough information (AAA proves similarity, not congruence)
7. Not enough information (SSA does not work)
8. $\overline{QR}$; $\angle R$
9. $x = 6$
10. SAS; $\overline{AC} \cong \overline{DF}$

## Station 3: Isosceles and Equilateral Triangles

1. $m\angle C = 50°,\ m\angle A = 80°$
2. $70°$ $\quad \frac{180 - 40}{2}$
3. $x = 10;\ 35°$ each; $110°$
4. $x = 4;\ P = 42$
5. $x = 5;\ AB = 31$
6. The two sides opposite the $55°$ angles
7. $x = 15;\ m\angle A = 100°$
8. $x = 8;\ 17,\ 17,\ 16$
9. 17
10. $45°$

## Station 4: Midsegments, Medians, and Special Segments

1. 7
2. 18
3. $x = 4;\ DE = 11,\ BC = 22$
4. 20
5. 12; 6
6. 24
7. $x = 6;\ PA = 23$
8. $x = 6;\ 21$
9. centroid; orthocenter; circumcenter; incenter
10. $(3, 3)$ *average the coordinates*